

Physics - 0 2026 Spring

Homework

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*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

Problem 1. (15pts) A small block with mass 0.04 kg slides in a vertical circle of radius $R = 0.5\text{ m}$ on the inside of a circular track. During one of the revolutions of the block, when the block is at the bottom of its path, point A , the normal force exerted on the block by the track has magnitude 3.95 N . In this same revolution, when the block reaches the top of its path, point B , the normal force exerted on the block has magnitude 0.68 N . How much work is done on the block by friction during the motion of the block from point A to point B ?

Solution. Denote the normal force and the speed of the block at point A and B as N_A and v_A , N_B and v_B respectively. The gravitational force acting on the block is given by mg , where m is the mass of the block and g is the acceleration due to gravity.

Then v_A and v_B is given by

$$v_A = \sqrt{\frac{R(N_A - mg)}{m}} \quad \text{and} \quad v_B = \sqrt{\frac{R(N_B + mg)}{m}}.$$

The decrease in kinetic energy of the block from point A to point B is $m(v_A^2 - v_B^2)/2$, and the decrease in potential energy is $2mgR$. Therefore, the work done by friction is given by

$$W = \frac{m(v_A^2 - v_B^2)}{2} + 2mgR.$$

Substituting the values of m , R , N_A , N_B , and g into the above equation, we got

$$W = 1.0175\text{ J}.$$

Problem 2. (15pts) Consider an object of mass m that is released from rest in a fluid with linear drag $\vec{F}_d = -b\vec{v}$. Compute the energy as a function of time and show that $\Delta E = E(t) - E(0)$ equals the work done by the drag force in time t .

Solution. Denote the velocity of the object at time t as $v(t)$. The equation of motion for the object can be written as

$$m \frac{dv}{dt} = mg - bv, \quad v(0) = 0.$$

Solving this differential equation, we get

$$v(t) = \frac{mg}{b}(1 - e^{-bt/m}).$$

Denote the distance traveled by the object at time t as $x(t)$. We can find $x(t)$ by integrating $v(t)$:

$$x(t) = \int_0^t v(t') dt' = \frac{mg}{b} \left(t + \frac{m}{b}(e^{-bt/m} - 1) \right).$$

Then we can work out that

$$\Delta E = -mgx(t) + \frac{1}{2}mv(t)^2 = \frac{m^3g^2}{2b^2} (e^{-2bt/m} - 4e^{-bt/m} + 3) - \frac{m^2g^2}{b}t.$$

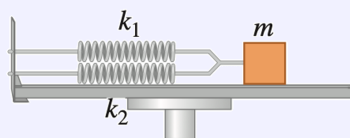
On the other hand, the work done by the drag force in time t is given by

$$W = \int_0^t F_d v(t') dt' = -b \int_0^t v(t')^2 dt' = \frac{m^3g^2}{2b^2} (e^{-2bt/m} - 4e^{-bt/m} + 3) - \frac{m^2g^2}{b}t.$$

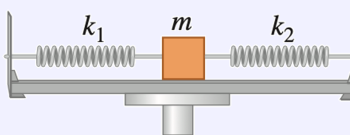
Hence, we have $\Delta E = E(t) - E(0) = W$, which shows that the change in energy of the object equals the work done by the drag force.

Problem 3. (40pts) Two springs with the same unstretched length but different spring constants k_1 and k_2 are attached to a block with mass m on a frictionless surface. Calculate the effective spring constant k_{eff} such that $\sum F = -k_{\text{eff}}x$ for each of the cases shown in the pictures below.

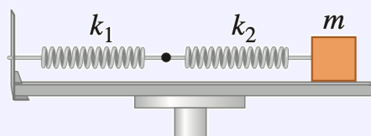
(a) (10pts)



(b) (10pts)



(c) (10pts)



(d) (10pts) Assume now that you have an object of mass m attached to only one spring with spring constant k and oscillating with frequency f_1 . If the spring is cut in half and the same object is attached to one of the two halves, its frequency is now f_2 . What is the ratio f_2/f_1 ?

Solution. (a) $F = F_1 + F_2 = (k_1 + k_2)x$, hence $k_{\text{eff}} = k_1 + k_2$.

(b) $F = F_1 - F_2 = k_1x - k_2(\text{const.} - x) = (k_1 + k_2)x + \text{const.}$, hence $k_{\text{eff}} = k_1 + k_2$.

(c) $x = x_1 + x_2 = F/k_1 + F/k_2$, hence $k_{\text{eff}} = k_1k_2/(k_1 + k_2)$.

(d) The frequency of oscillation is given by

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}.$$

When the spring is cut in half, the spring constant of each half becomes $2k$ because $k = (2k)^2/(2k + 2k)$.

Therefore, the new frequency f_2 is given by

$$f_2 = \frac{1}{2\pi} \sqrt{\frac{2k}{m}}.$$

The ratio of the frequencies is

$$\frac{f_2}{f_1} = \frac{\sqrt{2}}{1} = \sqrt{2}.$$

Problem 4. (30pts) Consider a one dimensional version of the “Mexican hat” potential, which is given by

$$U(x) = -\alpha x^2 + \beta x^4$$

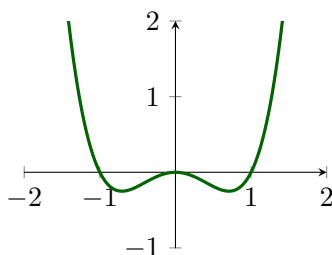
where $\alpha > 0$ and $\beta > 0$. (A higher-dimensional version of this potential is related to the “Higgs mechanism” that is responsible for giving particles a mass and ultimately led to the Nobel Prize in physics in 2013.)

(a) (10pts) Sketch the potential and determine the location of the local maximum and local minima.

(b) (10pts) If you release the particle from the local maximum (by giving it a small perturbation), what is its speed at the local minima? How far does the particle get before turning back?

(c) (10pts) Find the period of oscillation for a particle stuck around one of the local minima assuming that $x - x_0$ is small where x_0 is the location of the local minimum.

Solution. (a) The draft of the potential with $\alpha = \beta = 1$ is shown below. The local maximum is located at $x = 0$ and the local minima are located at $x = \pm\sqrt{\alpha/(2\beta)}$.



(b) Suppose the speed of the particle at the local minima is v . Since the particle is released from rest at the local maximum, the total energy of the system is given by $E = U(0) = 0$. According to the conservation of energy, we have

$$E = E_k + U(x_0) \implies 0 = \frac{1}{2}mv^2 - \alpha x_0^2 + \beta x_0^4,$$

where $x_0 = \pm\sqrt{\alpha/(2\beta)}$. Solving for v , we get

$$v = \frac{\alpha}{\sqrt{2m\beta}}.$$

Denote the turning point of the particle as x_{turn} . At the turning point, the potential energy equals the total energy, hence we have

$$U(x_{\text{turn}}) = 0 \implies -\alpha x_{\text{turn}}^2 + \beta x_{\text{turn}}^4 = 0 \implies x_{\text{turn}} = \sqrt{\alpha/\beta}.$$

(c) The period of oscillation near x_0 is given by

$$T = 2\pi\sqrt{\frac{m}{U''(x_0)}} = \pi\sqrt{\frac{m}{\alpha}}.$$